

Data Analysis and Visualization

PART III: Analyzing data, Interpreting the analyses, Communicating the results statistics

Module: Statistics

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- Analyzing quantitative data
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- Basic event relations and probability laws
- Conditional probability and independence
- Bayes rule
- Discrete and continuous random variables
- Random sampling and sampling distributions
- Central Limit Theorem
- Confidence interval

3 Lab 5

Revise week 4

Creating visualization dashboards

- Creating Dashboards with Plotly: `Plotly_Basics.ipynb`
- Creating Dashboards with Dash: `dash_interactivity.py` and `flight_delay.py`
- Dataset `airline_data.csv`

Descriptive statistics

- Data types
- Analyzing quantitative data: Measures of center, measures of spread, the shape of the data, outliers
- Inferential statistics and descriptive statistics
- Simpson's Paradox phenomenon

Utilizing data is the new way of the world

- The word "data" is defined as distinct pieces of information.
- Data can come in many forms: spreadsheets, text, video, databases, images, audio, etc.
- Data is used to understand, improve every facet of our lives
 - ▶ Early disease detection, Medicine
 - ▶ Social networks: connect and communicate with people
 - ▶ Insurance and Banking
 - ▶ Education
 - ▶ Agriculture
 - ▶ Automotive
 - ▶ Manufacturing,

Before collecting data, we usually start with a question, or many questions, that we'd like to answer. The purpose of data is to help us in answering these questions.

Data types

① Quantitative vs. Categorical Data

- ▶ Quantitative data takes on numeric values that allow us to perform mathematical operations (like the number of dogs). Quantitative data takes on numeric values that allow us to perform mathematical operations.
- ▶ Categorical are used to label a group or set of items (like dog breeds - Collies, Labs, Poodles, etc.).

② Categorical Ordinal vs. Categorical Nominal

- ▶ Categorical Ordinal data take on a ranked ordering (like a ranked interaction on a scale from Very Poor to Very Good with the dogs).
- ▶ Categorical Nominal data do not have an order or ranking (like the breeds of the dog).

Data types (cont.)

- ① Quantitative vs. Categorical Data
- ② Categorical Ordinal vs. Categorical Nominal
- ③ Continuous vs. Discrete
 - ▶ Continuous data can be split into smaller and smaller units.
Example: age of the dog - we can measure the units of the age in years, months, days, hours, seconds, but there are still smaller units that could be associated with the age.
 - ▶ Discrete data only takes on countable values. Example: The number of dogs we interact with.

To summarize, we have 2 main data types, each with two subgroups

- Quantitative data can be divided into continuous and discrete.
- Categorical can be split into nominal and ordinal.

Analyzing quantitative data

Four main aspects to analyzing quantitative data

- Measures of Center
- Measures of Spread
- The Shape of the data
- Outliers

Analyzing categorical data: usually look at the counts or proportion of individuals that fall into each group. Example: look at the breeds of the dogs, we'd care about how many dogs are of each breed, what proportion of dogs are of each breed type.

Three Measures of Center

- Mean
- Median
- Mode

Measures of center (Mean)

The mean is often called the average or the expected value in mathematics. We calculate the mean by adding all of our values together, and dividing by the number of values in our dataset. In mathematical notation:

- Assume that n is the total number of values in our dataset
- Each value of our dataset is denoted by x_i for $i = 1, \dots, n$
- We write $\bar{x}$ to represent the mean of a dataset

Then,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Measures of center (Median)

The median splits our data so that 50% of our values are lower and 50% are higher.

- Median for odd values: If we have an odd number of observations, the median is simply the number in the direct middle. Example, if we have 7 observations, the median is the fourth value when numbers are ordered from smallest to largest.
- Median for even values: If we have an even number of observations, the median is the average of the two values in the middle. Example, if we have 8 observations, we average the fourth and fifth values together when our numbers are ordered from smallest to largest.

In order to compute the median we **MUST** sort our values first.
Whether we use the mean or median to describe a dataset is largely dependent on the shape of our dataset and if there are any outliers.

Measures of center (Mode)

- The mode is the most frequently observed value in our dataset.
- There might be multiple modes for a particular dataset, or no mode at all.
 - ▶ No Mode if all observations in our dataset are observed with the same frequency. Example the dataset 1, 1, 2, 2, 3, 3, 4, 4
 - ▶ Many Modes if two (or more) numbers share the maximum value. Example the dataset 1, 2, 3, 3, 3, 4, 5, 6, 6, 6, 7, 8, 9 has two modes 3 and 6, because these values share the maximum frequencies at 3 times, while all other values only appear once.

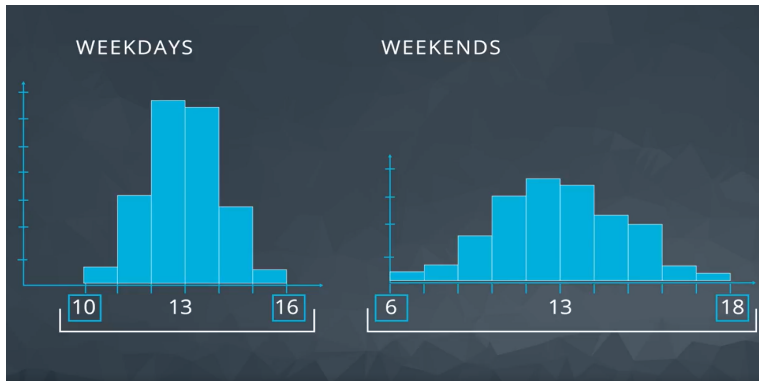
Measures of spread

Measures of Spread are used to provide us an idea of how spread out our data are from one another. Common measures of spread include:

- Range: the difference between the maximum (the largest value in the dataset) and the minimum (the smallest number in the dataset).
- Interquartile Range (IQR): the difference between Q_3 (the value such that 75% of the data fall below) and Q_1 (the value such that 25% of the data fall below). Recall Q_2 is the median of the dataset.
- Standard Deviation
- Variance

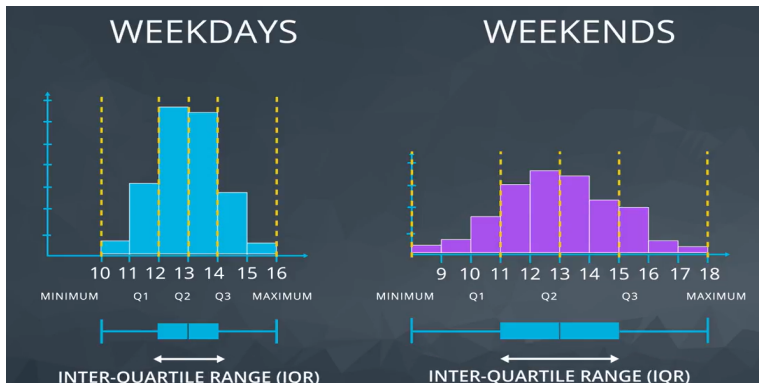
Histograms are super useful to understanding the different aspects of quantitative data:

- center
- spread
- shape
- outliers



Measures of spread (Interquartile Range)

Although histograms are preferred in most cases, a box plot can be useful for quickly comparing the spread of two data sets across some key metrics, like our quartiles, and the maximum and minimum.



What if we want just a single value to be able to compare the two distributions spreads?

Measures of spread (standard deviation & variance)

The variance is the average squared difference of each observation from the mean, which is calculated as

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

The standard deviation is equal to $\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$

It should be preferred to calculate variance of a dataset by dividing by $n - 1$ because dividing by n underestimates the variance

$$\frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Reference:

<https://www.youtube.com/watch?v=sHRBg6BhKjl&t=499s>

Measures of spread - Important points

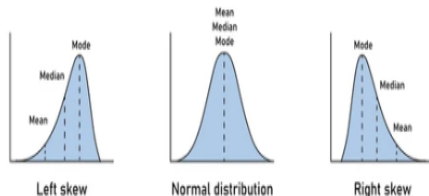
- The variance is used to compare the spread of two different groups. A set of data with higher variance is more spread out than a dataset with lower variance. Be careful, there might just be outlier(s) increasing the variance, when most of the data are actually very close.
- When comparing the spread between two datasets, the units of each must be the same.
- When data are related to money or the economy, higher variance (or std) is associated with higher risk.
- Standard deviation is used more often in practice than the variance, because it shares the units of the original dataset.

Use in the world: The std is associated with risk in finance, assists in determining the significance of drugs in medical studies, measures the error of results for predicting anything from the amount of rainfall we can expect tomorrow to your predicted commute time tomorrow.

Shape of the data

A histogram can quickly identify the shape of data. Distribution of data is frequently associated with one of the three shapes:

- Right-skewed: Mean greater than median. Examples: amount of drug remaining in a blood stream, time between phone calls at a call center, time until light bulb dies
- Left-skewed: Mean less than median. Examples: grades as a percentage in universities, age of death, asset price changes
- Symmetric (frequently normally distributed): Mean equals Median. Examples: height, weight, errors, precipitation



Outliers

- Outliers are points that fall very far from the rest of our data points. This influences measures like the mean and standard deviation much more than min, max, Q_1 , Q_2 , Q_3 .
- Different techniques for identifying outliers: e.g., “A Review and Comparison of Methods for Detecting Outliers” by Songwon Seo
- When outliers are present we should consider the following points:
 - ▶ Noting they exist and the impact on summary statistics.
 - ▶ If typo - remove or fix
 - ▶ Understanding why they exist, and the impact on questions we are trying to answer about our data.
 - ▶ Reporting the summary values (min, max, Q_1 , Q_2 , Q_3) is often a better indication than measures like the mean and standard deviation when we have outliers.
 - ▶ Be careful in reporting. Know how to ask the right questions.

Outliers advice

- Plot your data to identify if you have outliers.
- Handle outliers, e.g., methods from academic papers.
- If no outliers and your data follow a normal distribution - use the mean and standard deviation to describe your dataset, and report that the data are normally distributed. Check whether data are normally distributed by using normal quantile plots ([link](#)) and statistical methods like Kolmogorov-Smirnov test ([link](#))
- If you have skewed data or outliers, use the five number summary to summarize your data and report the outliers.

Look at histograms and box plots to visualize our quantitative data. Identifying outliers and the shape associated with the distribution of data are easier when using a visual as opposed to using summary statistics.

Inferential statistics and descriptive statistics

- ① Descriptive statistics: describe collected data by using the measures such as: measures of center, measures of spread, shape of our distribution, and outliers. Plots of data to gain a better understanding.
- ② Inferential statistics: use collected data to draw conclusions to a larger population. Performing inferential statistics well requires that we take a sample that accurately represents our population of interest. You look at specific examples to identify the
 - ▶ Population - our entire group of interest.
 - ▶ Parameter - numeric summary about a population
 - ▶ Sample - subset of the population
 - ▶ Statistic numeric summary about a sample

Many career paths involving Machine Learning and Artificial Intelligence are aimed at using collected data to draw conclusions about entire populations at an individual level.

Simpson's Paradox phenomenon

- Simpson's Paradox is a statistical phenomenon where a trend that appears in several different groups of data reverses or disappears when the groups are combined. Simpson's Paradox reminds us to be cautious when interpreting aggregated data. It shows the importance of considering confounding variables (like group membership) that might influence the observed outcome.
- Examples: Medical studies (treatment effectiveness), College admissions (gender bias), Sports statistics, Criminal justice and discrimination studies

Simpson's Paradox phenomenon

Example:

Suppose two treatments for a medical condition are being compared:

Group	Treatment A Success Rate	Treatment B Success Rate
Group 1	90%	80%
Group 2	75%	65%

In both groups, Treatment A looks better.

But when you combine both groups (say, because Group 1 is small in Treatment A but large in Treatment B), you might get:

Combined	Treatment A Success Rate	Treatment B Success Rate
All Groups	70%	75%

Now, **Treatment B appears better overall**, even though it was worse in every subgroup. That's Simpson's Paradox.

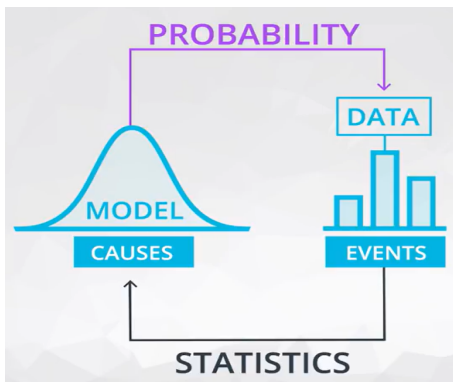
Probability

- Interpretation of probability
- Basic event relations and probability laws
- Conditional probability and independence
- Bayes rule
- Discrete and continuous random variables
- Random sampling and sampling distributions
- Central Limit Theorem
- Confidence interval

Recall the basic of probability

- In probability, we make predictions about future events based on models or causes that we assume
- In statistics we analyze the data from the past events to infer what those models or causes could be.

There's almost an opposite relation between proba and stat. In one you're predicting data and in the other you're using data to predict.



Classical Interpretation of Probability

- Originated from games of chance (e.g., coin tosses, card draws).
- Probability is calculated based on equally likely outcomes:

$$P(\text{event } E) = \frac{\text{number of outcomes } N_e \text{ favorable to event } E}{\text{total number of possible outcomes } N}$$

- Example:
 - ▶ Probability of heads in a fair coin flip = $1/2$
 - ▶ Probability of drawing an ace from a 52-card deck = $4/52$
- Assumes all outcomes are equally likely. If this assumption does not hold, the probabilities indicated by the classical interpretation of probability will be in error.

Relative frequency concept of probability

- Based on empirical data from repeated experiments.
- If event E occurs n_e times in n trials:

$$P(\text{event } E) \approx \frac{n_e}{n}$$

- More repetitions lead to more accurate probability estimates.
- Practical and testable interpretation.

We say “approximately” because we think of the actual probability $P(\text{event } E)$ as the relative frequency of the occurrence of event E over a very large number of observations or repetitions of the phenomenon. If an experiment is repeated a large number of times and event E occurs 30% of the time, then 0.30 should be a very good approximation to the probability of event E .

Personal/subjective interpretation of probability

- Used in one-shot or non-repetitive situations.
- Reflects personal belief or judgment.
- Example: Estimating the chance that a law will pass.
- Not verifiable or consistent across individuals.

Basic event relations and probability laws

Definition (Mutually exclusive events)

Two events A and B are said to be **mutually exclusive** if, when the experiment is performed a single time, the occurrence of one of the events excludes the possibility of the occurrence of the other. That is, events A and B have no outcomes in common:

$$A \cap B = \emptyset$$

Definition

If two events A and B are mutually exclusive, the probability that either event occurs is given by:

$$P(\text{either } A \text{ or } B) = P(A) + P(B)$$

Basic event relations and probability laws

Definition

The complement of an event A is the event that A does not occur. The complement of A is denoted by the symbol $\bar{A}$.

If A and B are any two mutually exclusive events associated with an experiment, then $P(A)$ and $P(B)$ must satisfy

- 1 $0 \leq P(A) \leq 1$ and $0 \leq P(B) \leq 1$. The probability of any event must be between 0 and 1, inclusive.
- 2 $P(\text{either } A \text{ or } B) = P(A) + P(B)$
- 3 $P(A) + P(\bar{A}) = 1$ and $P(B) + P(\bar{B}) = 1$. The sum of all possible events is equal to 1. The probability of the complement event is 1 minus the probability of an event.

Basic event relations and probability laws

Definition

The union of two events A and B is the set of all outcomes that are included in either A or B (or both). The union is denoted as $A \cup B$.

Definition

The intersection of two events A and B is the set of all outcomes that are included in both A and B . The intersection is denoted as $A \cap B$.

Definition

Consider two events A and B ; the probability of the union of A and B is given by:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional probability

Definition

Consider two events A and B with nonzero probabilities, $P(A)$ and $P(B)$. The **conditional probability** of event A , given event B , is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Definition

The **probability of the intersection** of two events A and B is

$$P(A \cap B) = P(B | A)P(A) = P(A | B)P(B)$$

Example:

$$P(\text{positive} | \text{disease}) = \frac{P(\text{positive} \cap \text{disease})}{P(\text{disease})}$$

Independent events

Definition

Two events A and B are independent events if

$$P(A | B) = P(A) \quad \text{or} \quad P(B | A) = P(B).$$

(If $P(A | B) = P(A)$, then $P(B | A) = P(B)$, and vice versa)

This leads to special case $P(A \cap B) = P(A)P(B) = P(A)P(B | A)$

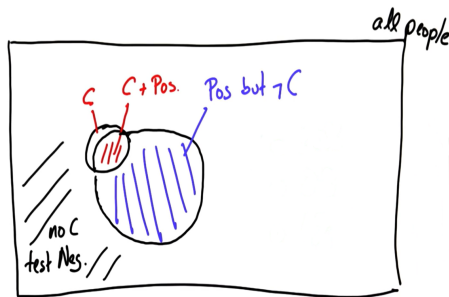
The concept of independence is important in sampling. Later, we will discuss drawing samples from two (or more) populations to compare the means, variances, or other population parameters. For most of these applications, we will select samples in such a way that the observed values in one sample are independent of the values that appear in another sample. We call these **independent samples**.

Bayes rule

Example of cancer:

- There's a specific cancer C that occurs in 1% of the population
- The test for this cancer and with 90% chance it is positive if they have this cancer C
- The test sometimes is positive, even if you don't have C . Let's say with another 90% chance it's negative if we don't have C

Question: Without further symptoms, you take the test, and the test comes back positive. What is probability of having that cancer?



Bayes rule

prior: $P(C) = 0.01 = 1\%$ $P(\neg C) = 0.99$
 $P(\text{Pos}|C) = 0.9 = 90\%$
 $P(\text{Neg}|\neg C) = 0.9$ $P(\text{Pos}|\neg C) = 0.1$

~~posterior~~
joint

$$P(C, \text{Pos.}) = P(C) \cdot P(\text{Pos}|C) = 0.009$$
$$P(\neg C, \text{Pos.}) = P(\neg C) \cdot P(\text{Pos}|\neg C) = 0.099$$

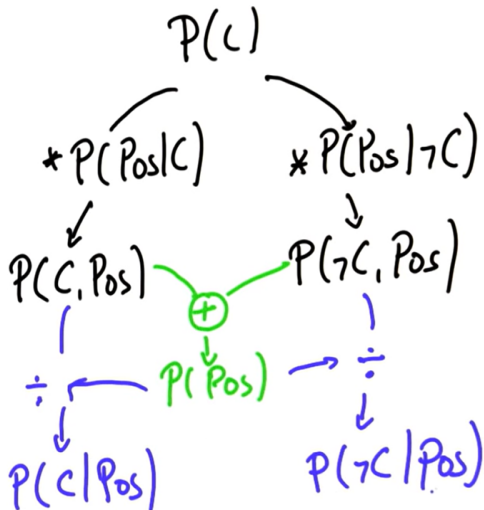
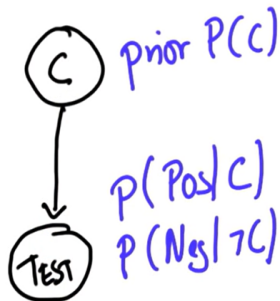
normalizer

$$P(\text{Pos}) = P(C, \text{Pos}) + P(\neg C, \text{Pos}) = 0.108$$

posterior:

$$P(C|\text{Pos}) = \boxed{0.0833}$$
$$P(\neg C|\text{Pos}) = \boxed{0.9167}$$

Bayes rule



Bayes' formula

Bayes' Formula:

If A and B are any events whose probabilities are not 0 or 1, then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

Bayes' Formula:

If $A_1, \dots, A_k$ are mutually exclusive events, and if $B_1, \dots, B_m$ are m possible, mutually exclusive, observable events, then

$$P(A_i|B_j) = \frac{P(A_i \cap B_j)}{P(B_j)} = \frac{P(B_j|A_i)P(A_i)}{\sum_{i=1}^k P(B_j|A_i)P(A_i)}$$

Prior $P(B_j|A_i)$ and posterior $P(A_i|B_j)$ refer to probabilities before and after observing an event B_j .

Random variables

Definition

When observations on a quantitative random variable can assume only a countable number of values, the variable is called a discrete random variable.

Examples: Change in the number of accidents per month at an intersection after a new signaling device has been installed.

Definition

When observations on a quantitative random variable can assume any one of the uncountable number of values in a line interval, the variable is called a continuous random variable.

Examples: temperature, pressure, height, weight, and distance
The distinction between discrete and continuous r.v is pertinent when seeking the probabilities associated with specific values of a r.v.

Probability distributions for discrete r.vs

1- Binomial distribution: helps us determine the probability of a string of independent 'coin flip like events'.

- The probability mass function of a Binomial distribution, denoted $X \sim B(n, p)$, is

$$P(X = k) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

where n is the number of events, k is the number of "successes", and p is the probability of "success".

- Examples: calculate probability of 3 heads occurring in 10 flips, probability of observing 8 or more heads occurring in 100 flips, probability of not observing any heads in 20 flips.
- Mean μ and Standard Deviation σ :

$$\mu = np \text{ and } \sigma = \sqrt{np(1-p)}$$

Probability distributions for discrete r.vs

2- Poisson distribution: Let y be the number of events occurring during a fixed time interval of length t or a fixed region R of area or volume $m(R)$. Then the probability distribution of y is Poisson, provided certain conditions are satisfied:

- Events occur one at a time; two or more events do not occur precisely at the same time or in the same space.
- The occurrence of an event in a given period of time or region of space is independent of the occurrence of the event in a nonoverlapping time period or region of space; that is, the occurrence (or nonoccurrence) of an event during one period or in one region does not affect the probability of an event occurring at some other time or in some other region.
- The expected number of events during one period or in one region μ is the same as the expected number of events in any other period or region.

Probability distributions for discrete r.vs

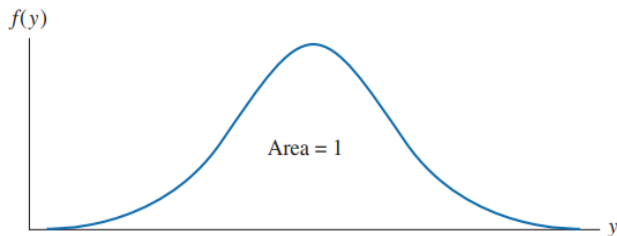
2- Poisson distribution:

- Example: the number of arrivals of customers at a checkout counter, parking lot toll booth, inspection station, or garage repair shop during a specified time interval can often be modeled by a Poisson distribution.
- The Poisson probability of observing y events in a unit of time or space is given by the formula

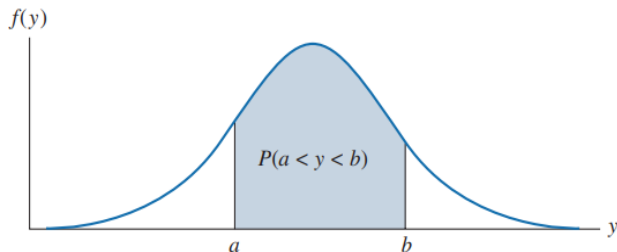
$$P(y) = \frac{\mu^y e^{-\mu}}{y!}$$

where e is equal to 2.71828.

Probability distributions for continuous r.vs



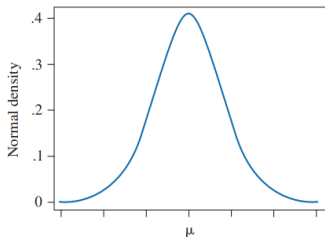
(a) Total area under the curve



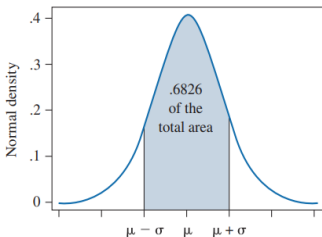
(b) Probability

Probability distributions for continuous r.vs

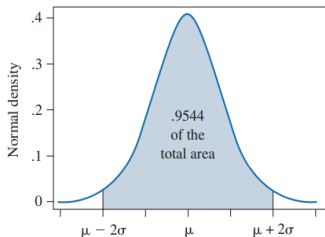
Normal distribution



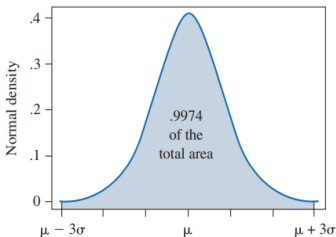
(a) Normal curve



(b) Area under normal curve within 1 standard deviation of mean



(c) Area under normal curve within 2 standard deviations of mean



(d) Area under normal curve within 3 standard deviations of mean

Normal distribution

We can use the normal approximation to the binomial distribution when $np \geq 10$ and $n(1 - p) \geq 10$, where n is the number of trials, p is the probability of success, $1 - p$ is the probability of failure. Particularly, if $X \sim B(n, p)$ then

$$X \approx N(\mu, \sigma^2)$$

where $\mu = np$ and $\sigma^2 = np(1 - p)$

Note that the normal approximation is quite good when np and $n(1 - p)$ exceed about 20. In the middle zone, np or $n(1 - p)$ between 5 and 20, a modification called a continuity correction makes a substantial contribution to the quality of the approximation.

Normal distribution

Example: Let $X \sim \text{Bin}(100, 0.4)$. Then,

$$\mu = 100 \cdot 0.4 = 40, \quad \sigma = \sqrt{100 \cdot 0.4 \cdot 0.6} = \sqrt{24} \approx 4.90$$

To approximate $P(X \leq 45)$, apply the continuity correction:

$$P(X \leq 45) \approx P(Y \leq 45.5)$$

Standardize using the Z -score:

$$Z = \frac{45.5 - 40}{4.90} \approx 1.12$$

Using standard normal tables:

$$P(Z \leq 1.12) \approx 0.8686$$

Random sampling and sampling distributions

Definition

A sample of n measurements selected from a population is said to be a random sample if every different sample of size n from the population has an equal probability of being selected.

Sample selection bias exists whenever there is a systematic tendency to overrepresent or underrepresent some part of the population.

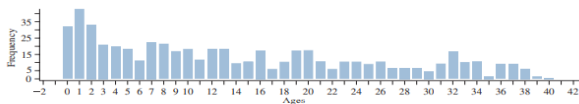
Example, a survey of households conducted during the week entirely between the hours of 9 a.m. and 5 p.m. would be severely biased toward households with at least one member at home. Hence, any inferences made from the sample data would be biased toward the attributes or opinions of those families with at least one member at home and may not be truly representative of the population of households in the region.

Random sampling and sampling distributions

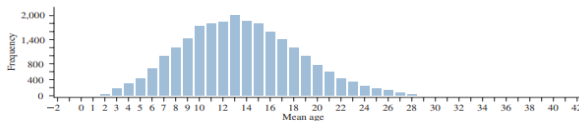
- Numerical descriptive measures of a population (parameters): example, the mean μ and the standard deviation σ of a population are parameters
- Numerical descriptive measures of a sample (statistics): example, the sample mean $\bar{y}$ and sample standard deviation s are statistics.

A sample statistic has a probability distribution. We call the probability distribution of a sample statistic the sampling distribution of that statistic. Stated differently, the sampling distribution of a statistic is the population of all possible values for that statistic.

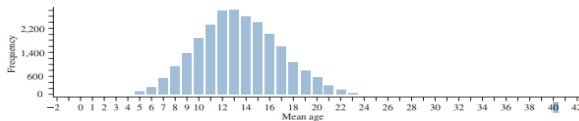
Sampling distributions



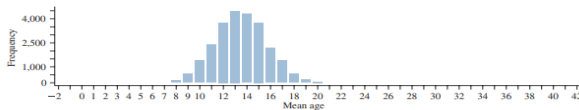
(a) Histogram of ages for 500 pennies



(b) Sampling distribution of $\bar{y}$ for $n = 5$



(c) Sampling distribution of $\bar{y}$ for $n = 10$



(d) Sampling distribution of $\bar{y}$ for $n = 25$

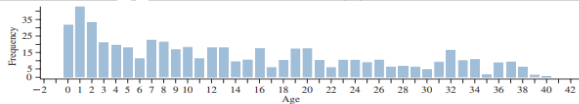
Sampling distributions

The sampling distribution of $\bar{y}$ has mean $\mu_{\bar{y}}$ and standard deviation $\sigma_{\bar{y}}$, which are related to the population mean μ and standard deviation σ by the following relationships (central limit theorem said):

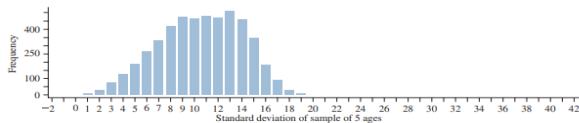
$$\mu_{\bar{y}} = \mu \text{ and } \sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$$

The quantity $\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$ is called **the standard error** of $\bar{y}$.

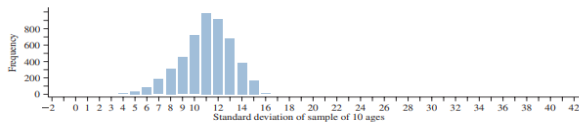
Sampling distributions



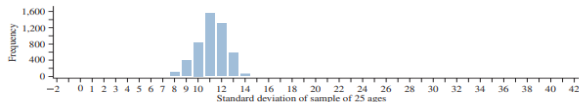
(a) Histogram of ages for 500 pennies



(b) Sampling distribution of standard deviation for $n = 5$



(c) Sampling distribution of standard deviation for $n = 10$



(d) Sampling distribution of standard deviation for $n = 25$

Central Limit Theorem for sample mean

Let $\bar{y}$ denote the sample mean computed from a random sample of n measurements from a population having a mean μ and finite standard deviation σ . Let $\mu_{\bar{y}}$ and $\sigma_{\bar{y}}$ denote the mean and standard deviation of the sampling distribution of $\bar{y}$, respectively. Based on repeated random samples of size n from the population, we can conclude the following:

- $\mu_{\bar{y}} = \mu$
- $\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$
- When n is large, the sampling distribution of $\bar{y}$ will be approximately normal (with the approximation becoming more precise as n increases).
- When the population distribution is normal, the sampling distribution of $\bar{y}$ is exactly normal for any sample size n .

Central Limit Theorem for sample sum

Let Σy denote the sum of a random sample of n measurements from a population having a mean μ and finite standard deviation σ . Let $\mu_{\Sigma y}$ and $\sigma_{\Sigma y}$ denote the mean and standard deviation of the sampling distribution of Σy , respectively. Based on repeated random samples of size n from the population, we can conclude the following:

- $\mu_{\Sigma y} = n\mu$
- $\sigma_{\Sigma y} = \sqrt{n}\sigma$
- When n is large, the sampling distribution of Σy will be approximately normal (with the approximation becoming more precise as n increases).
- When the population distribution is normal, the sampling distribution of Σy is exactly normal for any sample size n .

Central Limit Theorem

- Notice: It suggests that the Central Limit Theorem holds for the sample size $n > 30$. Any inference based on the normality of $\bar{y}$ for the sample size $n \leq 30$ should be examined carefully (example, histogram plots at first to look at datasets).
- In practice, a sample is taken only once, and only one value of the sample statistic is calculated. A sampling distribution is not s.th you can see in practice; it is not an empirically observed distribution. Rather, it is a theoretical concept, a set of probabilities derived from assumptions about the population and about the sampling method.

Confidence interval

Revise by yourself. Reference links:

- [Confidence Intervals, Clearly Explained \(Click\)](#)
- [Bootstrapping Main Ideas \(Click\)](#)

Lab 5

- Doing exercises about probability and statistics.
- Apply the probability and statistic theory to figure out the solutions of your own research.